SHIVANSHU PANDEY

M.Sc. PHYSICS | IIT HYDERABAD

shivanshu.sp125@gmail.com | +91 9918295630 | github.com/sh1vanshupandey

Astrophysics · Computational Physics · Scientific Data Analysis

EDUCATION

Indian Institute of Technology Hyderabad | 2025–2027

Master of Science, Physics — currently in 3rd semester

Core courses: High Energy Particle Astrophysics, Plasma Physics, Particle Physics, Classical Mechanics, Quantum Mechanics I & II, Electrodynamics, Mathematical Physics I & II, Statistical Mechanics, Computational Physics, Optics & Photonics, Atomic & Molecular Physics.

VBS Purvanchal University | 2020–2023

Bachelor of Science — Physics, Chemistry, Mathematics

Core courses: Calculus, Linear Algebra, Differential Equations, Real & Complex Analysis, Numerical Analysis & C Programming, Optics & Lasers, Electronics, Solid State & Nuclear Physics.

SELECTED PROJECTS

Cosmic Ray Propagation in the Galaxy

- Developed 1D, 2D, and 3D simulations of cosmic-ray transport from supernova sources through the interstellar medium.
- Used Monte Carlo methods for pitch-angle scattering and random walks of 1500+ particles.
- Solved the diffusion PDE with Crank–Nicolson and the Thomas algorithm; compared results statistically with analytical solutions.
- Applied Dirichlet boundary conditions to study galactic escape and particle residence times.

Quantifying Sunspots using SAOImage DS9

- Analysed solar FITS images to identify sunspots, measure their sizes, and estimate physical areas using image analysis.
- Gained experience in astronomical and observational data analysis.

Optical Spectrometer

- Designed and constructed a low-cost diffraction-grating spectrometer with webcam detection, slit, light-isolation enclosure, and alignment setup.
- Captured and analysed spectral data; studied wavelength separation and light-source intensity; estimated the Sun's temperature using Wien's displacement law.

8-bit Breadboard Computer

- Designed and tested logic gates, registers, counters, and memory circuits to explore computer architecture and instruction execution.

POSITIONS OF RESPONSIBILITY

- **Class Representative, M.Sc. Physics, IIT Hyderabad** | Jul 2025–Present
- **Core Member, Cepheid Astronomy Club** | Jun 2026–Present
- **Core Member, Nakshatra Astronomy Club** | Aug 2026–Present
- Volunteer, National Science Week, IIT Hyderabad
- Volunteer, National Space Week, IIT Hyderabad

ACHIEVEMENTS

- **IIT JAM Physics 2025:** Qualified, All India Rank 514.
- Shortlisted for iPhD programme interviews at IISER Pune, IISER Bhopal, and NISER Bhubaneswar.

EXPERIENCE & TRAINING

Geo-processing Using Python — IIRS (ISRO)

July 2023 · 12 hours

Completed training in Python-based geospatial data processing and remote sensing, including Earth-observation data analysis and geoprocessing techniques.

TECHNICAL SKILLS

Programming: Python

Scientific libraries: NumPy, Matplotlib, SciPy, Pandas

Astronomy software: SAOImage DS9, TOPCAT

Tools: GitHub, VS Code

Documentation: LaTeX

SCIENTIFIC INTERESTS

Computational astrophysics, cosmic-ray transport, cosmology, observational astronomy, statistical data analysis, numerical methods, optics, and scientific computing.